Microstructure and in vitro beta carotene bioaccessibility of heat processed orange fleshed sweet potato.

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Orange fleshed sweet potato (OFSP) has been identified as a good source of beta-carotene but the beta-carotene bioaccessibility is affected by processing. In this study, the effect of traditional heat processing methods on the microstructure and in vitro bioaccessibility of beta-carotene from OFSP were investigated. Bioaccessibility was determined using simulated in vitro digestion model followed by membrane filtration to separate the micellar fraction containing bioaccessible beta-carotene. Processing led to decrease in the amount of all-trans-beta-carotene and increase in 13-cis-beta-carotene. Processed OFSP had significantly higher ($P < 0.05$) bioaccessible beta-carotene compared to the raw forms. Bioaccessibility varied with processing treatments in the order; raw < baked < steamed/boiled < deep fried. Light microscopy showed that the microstructure of OFSP was disrupted by the processing methods employed. The cell walls of OFSP were sloughed by the traditional heat processing methods applied. The findings show that heat processing improves bioaccessibility of beta-carotene in OFSP and this was probably due to disruption of the tissue microstructure.